



YEAR 7 MATHEMATICS · CHAPTER 10

Interpreting graphs

Lesson 1 of 8

In this lesson you will learn to:

- read and interpret picture, column, sector, line and divided bar graphs
- explain what a graph tells us about real data

Name _____

Class _____

Teacher _____

Before you start

You will need

Pencil

Ruler

Calculator

Syllabus: MA4-DAT-C-01: classifies and displays data using a variety of graphical representations

WE DO

Grey boxes. Work with your teacher. Copy what they write on the board.

YOU DO

Blue boxes. Work on your own. Show your working in the box.

- 1 Work through **Level 1**, then **Level 2**, then **Level 3**. Try the **extension** if you finish early.

- 2 At the end, **show off your skill**: answer one exit ticket question, tear it off and hand it in.

Lesson map

Page	What is on it	Done
3	Learning intentions, key terms, notes and Level 1 teacher examples	☐
4	Level 1: YOU DO practice questions	☐
5	Level 2: WE DO teacher examples and YOU DO practice questions	☐
6	Level 2: YOU DO practice questions	☐
7	Level 3: WE DO teacher examples and YOU DO practice questions	☐
8	Level 3: YOU DO practice questions	☐
9	Extension questions and the tear-off exit ticket	☐
10	Lesson summary: a worked example for every level	☐

YOU DO

On your own: Warm-up. These skills will help you in this lesson. Show your working.

- 1 $\frac{1}{2}$ of 80

- 3 10% of 300

- 2 $\frac{1}{4}$ of 60

- 4 20 out of 100 as a percentage



Acknowledgement of artwork

The artwork used throughout this booklet series was created by **Marni Tuala**. We thank Marni for sharing this artwork with our school community.

Acknowledgement of Country

We acknowledge the Bundjalung people, the Traditional Custodians of the land on which Kingscliff High School stands, and pay our respects to Elders past and present.

LESSON 10.01

Interpreting graphs

Date _____

Learning intentions

We are learning to:

- read and interpret picture, column, sector, line and divided bar graphs
- explain what a graph tells us about real data

Success criteria

I can:

- ☐ use a **key** and a **scale** to read values from a graph
- ☐ estimate amounts from a **sector graph** using fractions of a circle
- ☐ read percentages from a **divided bar graph** and give reasons for patterns

Key terms Write the definition as your teacher explains it.

Key
Sector graph
Divided bar graph

Notes

Level 1 · Getting started

Books borrowed from the library

Monday	
Tuesday	
Wednesday	
Thursday	
Friday	

 = 4 books

WE DO

With your teacher: copy what they write on the board.

Example 1

How many books were borrowed on Tuesday?

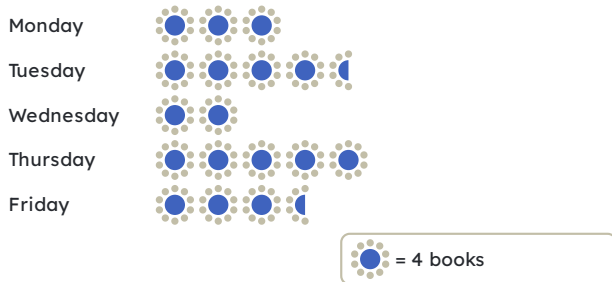
Example 2

How many more books on Thursday than Wednesday?

YOU DO

On your own: Set A. Use the library picture graph.

Books borrowed from the library

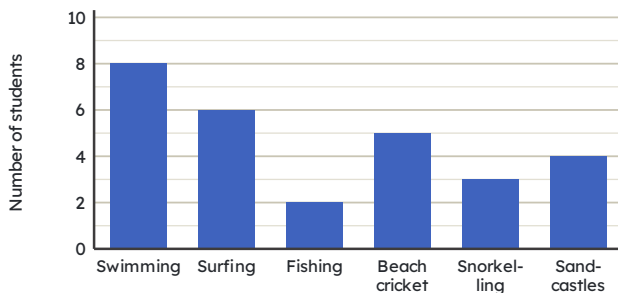


- 1 What does 1 symbol stand for?
- 2 What does $\frac{1}{2}$ a symbol stand for?
- 3 How many books on Monday?
- 4 How many books on Wednesday?
- 5 How many books on Friday?
- 6 Which day had the most books?
- 7 Which day had the fewest books?
- 8 How many more on Friday than Monday?
- 9 How many on Monday and Tuesday?
- 10 How many symbols show 16 books?

YOU DO

On your own: Set B. Use the column graph.

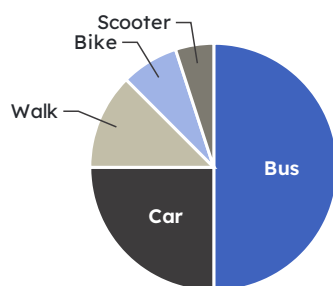
Favourite beach activity of 7K



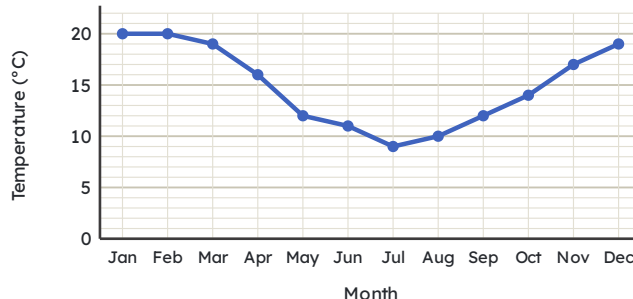
- 1 How many chose surfing?
- 2 How many chose fishing?
- 3 Which was the most popular?
- 4 Which was the least popular?
- 5 Which activity had 5 votes?
- 6 How many chose snorkelling?
- 7 How many more swam than fished?
- 8 How many chose surfing or fishing?
- 9 How many students are in 7K?
- 10 What does 1 gridline stand for?

Level 2 · Building up

How 80 Year 7 students travel to school



Mean minimum temperature, Kingscliff



WE DO With your teacher: copy what they write on the board.

Example 3 What fraction of the students catch the bus? How many students is that?

Example 4 Which month is coldest? How much colder is June than March?

YOU DO On your own: Set C. Questions 1–4 use the sector graph above.

1 Which way of travel is the most common?

3 Estimate how many students walk.

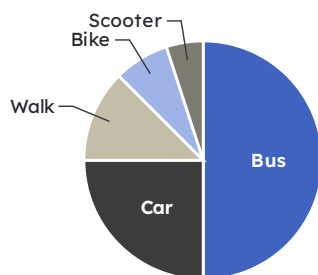
2 Estimate how many students travel by car.

4 True or false: fewer than 10 ride a bike.

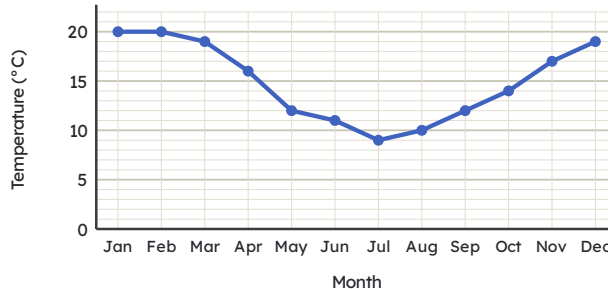
YOU DO

On your own: Set C continued. Question 5 uses the sector graph and 6–10 use the line graph.

How 80 Year 7 students travel to school



Mean minimum temperature, Kingscliff



- 5 How many students do **not** catch the bus?

- 8 Which month has a mean minimum of 14 °C?

- 6 Which 2 months are warmest? What is their mean minimum?

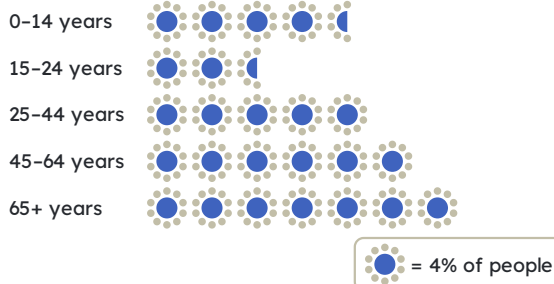
- 9 Which 2 months in a row have the biggest rise?

- 7 Name 2 months with the same mean minimum.

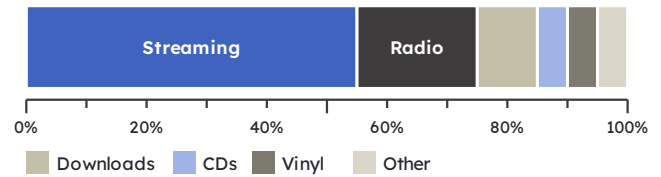
- 10 Which 2 months in a row have the biggest drop?

Level 3 · Challenge

Ages of people in a coastal town



How Australians listen to music



WE DO With your teacher: copy what they write on the board.

Example 5 What percentage of the town is 45+?
How many is that out of 5000?

Example 6 What % stream music? How might this look different 20 years ago?

YOU DO On your own: Set D. Use the picture graph above.

1 What percentage of people are aged 65+?

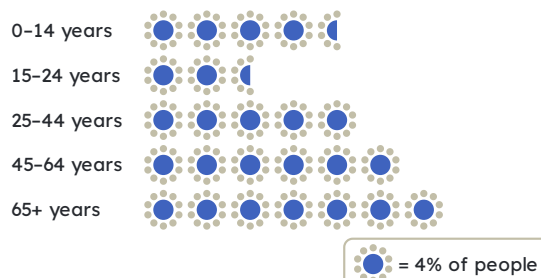
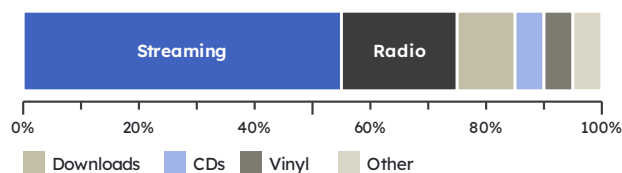
3 Town of 5000 people: how many are 0-14?

2 What percentage of people are under 25?

4 How many symbols would show 30% of the town?

YOU DO

On your own: Set D continued. Question 5 uses the picture graph and 6–10 use the divided bar graph.

Ages of people in a coastal town

How Australians listen to music


5 Why might 65+ be the biggest group in this town?

8 Survey of 400: how many use downloads?

6 Which 2 ways of listening add up to 75% of people?

9 Of 400: how many more stream than radio?

7 What fraction of people listen to the radio?

10 How might this graph change in 10 years? Why?

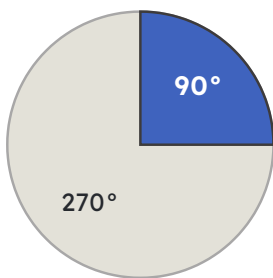
Extension · Stretch yourself

Finished Level 3? Stretch yourself here.

Worked example

In a survey of 72 students, the **Soccer** sector of a sector graph has an angle of 90° . How many students chose soccer?

1. A full circle is 360° , so 90° is $90/360 = \frac{1}{4}$ of the circle
2. $\frac{1}{4}$ of 72 = $72 \div 4 = 18$



90° out of 360°
= $\frac{1}{4}$ of the circle
= $\frac{1}{4}$ of the students

Answer: 18 students chose soccer

YOU DO

On your own: read the worked example, then try these.

- E1** A survey of 240 people has a **Tennis** sector of 60° . How many chose tennis?

- E2** In a survey of 180 people, 45 chose pizza. What angle should the pizza sector be?

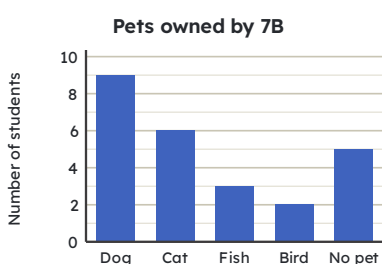
- E3** One symbol = 8 students. How many students do $3\frac{3}{4}$ symbols show? How would you show 20 students?

✂ Show off your skill: tear along this line and hand it to your teacher ✂

YOU DO Show off your skill

Choose **ONE** question. Show your working.

Name _____ Class _____



Level 1

How many students in 7B own a cat?

Level 2

What fraction of the class owns a dog?

Level 3

What % of 7B has no pet? How many of 200 Year 7s would you expect to have no pet?

Lesson summary

Missed the lesson? Start here.

Reading any graph

1. Read the **title**: what is the graph about?
2. Check the **key** or the **scale**.
3. Use a ruler to line up bars and points with the scale.
4. Answer with the right **units** (students, °C, %).

Level 1

In a picture graph,  = 10 students. A row shows 3½ symbols. How many students is that?

1. 3 full symbols = $3 \times 10 = 30$
2. Half a symbol = $10 \div 2 = 5$
3. $30 + 5 = 35$

Row

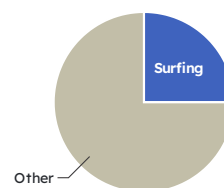

 = 10 students

Answer: 35 students

Level 2

A sector graph shows 80 people. The **Surfing** sector is a quarter of the circle. How many chose surfing?

1. A quarter means divide by 4
2. $80 \div 4 = 20$

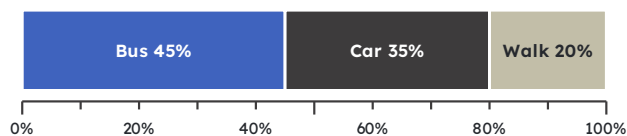


Answer: 20 people

Level 3

A divided bar graph shows 45% of 400 students catch the bus. How many students is that?

1. 45% means 45 out of every 100
2. 400 has 4 hundreds, so $45 \times 4 = 180$

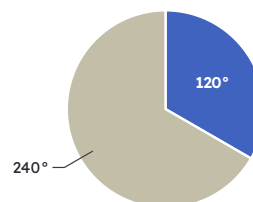


Answer: 180 students

Extension

A 120° sector shows the results for 90 people. How many people is that?

1. $120/360 = \frac{1}{3}$ of the circle
2. $\frac{1}{3}$ of 90 = $90 \div 3 = 30$



Answer: 30 people

This space is left blank on purpose. It is the back of your exit ticket.